MIL-D-37775B
19 June 1986
Superseding
MIL-D-37775A
19 September 1978

MILITARY SPECIFICATION

DRILL, PULP CANAL, DENTAL (Angle Handpiece)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers drills suitable for use in endodontic procedures.

1.2 Classification. The drills covered by this specification shall be of the following styles and sizes, as specified (see 6.1):

STYLE I. Gates Glidden

Size	Diameter
1	0.50 mm
2	0.70 mm
3	0.90 mm
4	1.10 mm
5	1.30 mm
6	1.50 mm

STYLE II. Peeso

Size	Diameter
4	1.30 mm
5	1.50 mm

2. APPLICABLE DOCUMENTS

2.1 Government documents.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-566 - Boxes, Folding Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Military

MIL-P-116 - Preservation, Methods of.

STANDARDS

Federal

FED-STD-151 - Metals, Test Methods.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

DRAWINGS

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

23309A - DRILL, PULP CANAL, DENTAL, Gates Glidden, Angle Handpiece.

23309B - DRILL, PULP CANAL, DENTAL, Peeso, Angle Handpiece.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 370 - Standard Methods and Definitions for Mechanical Testing of Steel Products.

ASTM E 92 - Standard Test Method for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Phila., PA. 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The Gates-Glidden drills shall be fabricated from a chromium-molybdenum-manganese stainless steel; the Peeso drills shall be fabricated from tungsten tool steel, having the following compositions:

Gates Glidden (Style I)		Peeso (Style II)	
Element	Percent	Element	Percent
Carbon	.20 - .25	Carbon	0.95 to 1.40
Silicon	.50 - .70	Manganese	0.25 to 0.40
Manganese	1.50 -1.85	Silicon	0.10 to 0.25
Phosphorus	Max 0.030	Chromium	0.20 to 0.40
Sulfur	0.018-0.25	Tungsten	0.70 to 2.00
Chromium	12.00-14.00	Vanadium	0.05 to 0.15
Nickel	.75 - 1.15	Lead	0.15 to 0.25
Molybdenum	1.10 - 1.30		

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3.2 Style, design, and dimensions. The drills shall be suitable for the enlargement, shaping, and debridement of the dental pulp chamber and canal when performing endodontic treatment. The drills shall be the Gates Glidden drills, Style I, sizes 1 through 6, and Peeso drills, Style II, sizes 4 and 5, suitable for use in the latch contra-angle handpieces. The drills shall be of one-piece construction, consisting of a shank, an elongated neck, and a drill head (operative end). Style I, Gates Glidden drills, shall conform to Defense Personnel Support Center (DPSC) Drawing 23309A, for sizes 1 through 6. Style II, Peeso drills, shall conform to Defense Personnel Support Center (DPSC) Drawing 23309B, for sizes 4 and 5.

3.2.1 Drawings. The drawings forming a part of this specification are not production drawings. Drawings establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the drawings to provide a product with proper design, configuration, fit, and function. No deviations from the dimensional limitations prescribed on the drawings are permissible without prior approval of the contracting officer.

3.3 Hardness. The operative end and neck of Style I drills shall have a Vickers hardness (Diamond pyramid hardness) of 500 to 680. The operative end cutting edges of the Style II drills shall have a Vickers hardness of 700 to 850. The neck of the Style II drills shall have a Vickers hardness of 380 to 530. The hardness shall be determined in accordance with the requirements of 4.3.2.

3.4 Tensile strength. The tensile strength of the raw drill wire for the Style I drills shall be 80 to 110 kilogram per square millimeter. The tensile strength of the raw drill wire for the Style II drills shall be 69 to 90 kilograms per square millimeter. Tensile strength shall be tested as specified in 4.3.3.

3.5 Sterilization. The drill shall show no corrosion and shall meet the hardness specified in 3.3, after having been subjected to the sterilization methods as specified in 4.3.4.

3.6 Runout. The total indicated runout of the neck of the drill shall not exceed 0.003 inch, when tested as specified in 4.3.5.

3.7 Marking. The shank of the drill shall be permanently and legibly marked to identify the size of the drill, such as a series of circular grooves corresponding to the size of the drill. Marking with manufacturer's name or trademark is optional.

3.8 Finish. The blade edge of the operative end shall have a sharp honed finish. The neck and latch end shall have a black finish, except for the shank portion, which shall have a bright finely machined finish.

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3.9 Workmanship. The drills shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. Drills shall be examined for defects including, but not limited to, those listed in table I.

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TABLE I. Classification of defects.

Categories	Defects
Major A:	
101	Style and size of drill not as specified.
102	Design and construction of drill not as specified.
103	Finish (fine machined and properly sharpened) of operative end of drill not as specified.
104	Latch portion of drill not cleanly machined or not free of burrs.
105	Shank of drill not smoothly finished.
Major B:	
151	Dimensions of latch end of shank not as specified.
152	Diameter of shank not as specified.
153	Blade end diameter not as specified.
154	Blade end length not as specified.
Minor:	
201	Dimensions of drill, other than Major B, not as specified.

The inspection levels and acceptable quality levels (AQLs) shall be as listed in table II. The unit of product for examination shall be one drill.

TABLE II. Examination criteria

Type defect	Inspection Level	AQL (percent defective)
Major A	II	1.0
Major B	S-1	1.5
Minor	II	2.5

4.2.2 Testing. Drills shall be tested as specified in table III and 4.3. Inspection levels and acceptable quality levels (AQLs) shall be as listed in table III. The unit of product for all tests, except chemical analysis and tensile strength, shall be one drill.

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TABLE III. Test criteria.

Characteristic	Component	Requirement	Test Procedure	Unit of Product	Inspection level	AQL (Percent defective)
Chemical analysis	Raw material	3.1	4.3.1	Lot of raw material	**	*
Hardness	Operative cutting edges	3.3	4.3.2	Drill	S-1	1.0
	Neck	3.3	4.3.2	Drill	S-1	1.0
Tensile strength	Steel stock	3.4	4.3.3	Lot of raw drill wire	***	*
Sterilization	Drill	3.5	4.3.4	Drill	S-1	1.0
Runout	Neck	3.6	4.3.5	Drill	S-4	*

* Acceptance number shall be zero for all sample sizes.

** A sufficient quantity of each lot of steel used in the fabrication of the drills shall be selected for chemical analysis.

*** A minimum of three specimens from each lot of raw drill wire shall be tested.

4.3 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.3.1 Chemical analysis. In the absence of certificates of quality for the steel, chemical analysis shall be conducted in accordance with method 111 or 112 of FED-STD-151.

4.3.2 Hardness. Testing for hardness shall be conducted in accordance with American Society for Testing and Materials (ASTM) Designation E 92 for Vickers hardness.

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4.3.3 Tensile strength. Tensile strength testing shall be performed in accordance with ASTM Designation A 370.

4.3.4 Sterilization. The test drills shall be cleaned in accordance with the instructions supplied by the manufacturer. The drills shall then be sterilized by one of the methods listed below for 3 cycles.

a. Dry Heat Sterilization - 121°C, 2 hours
b. Steam Autoclave - 121°C, 15 psi, 30 minutes
c. Unsaturated Chemical Vapor - 132°C, 20 psi, 20 minutes

After the drills are removed from the sterilizer, the entire surface area of the drill shall be examined for evidence of corrosion, and tested for hardness.

4.3.5 Runout. The drill shall be placed and held in a "V" block or precision chuck and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, a shadow-graph, or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the drill, and so positioned that the foot contacts the drill neck at the point nearest the head. The maximum and minimum readings obtained during the complete 360-degree rotation of the drill shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

4.4 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Drills shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Drills shall be dried in accordance with MIL-P- 116.

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5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit package. Six drills of one style and size only shall be packaged in a plastic container. The container shall consist of a plastic tray designed with six cavities. The cavities shall be molded in such a manner as to hold each drill securely in place and prevent damage to the operative end. The tray shall be designed with a clear plastic friction or slipover cover. The cover shall be secured with a clear tape. Each unit shall be further packaged in accordance with MIL-P-116, method 1C-1.

5.1.1.5 Intermediate package. Twelve unit packages containing drills of one style and size only shall be packaged in a box constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification. As an alternate, the intermediate package may be a plastic box having a friction closure. The closure shall be secured with tape.

5.1.2 Level C. Six drills of one style and size only, packaged in a plastic container as specified in 5.1.1.4 shall be further packaged level C conforming to the MIL-STD-794 requirements for the level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Fifty intermediate packages containing drills of one style and size shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping, and waterproofing shall be as specified in the box specification.

5.2.2 Level B. Fifty intermediate packages containing drills of one style and size shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. The level C packing for drills shall conform to MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quanitites less than 40 cubic feet need not be unitized. Pallet loads, including the pallet, shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Drills packed as specified in 5.2.1 and 5.2.2, shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Drills, packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B, and C. Each unit package, intermediate and exterior container, and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.4.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Style and size desired (see 1.2).
c. Selection and applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 National Stock Number (NSN) coverage. This specification covers the following items listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-01-011-2916	DRILL, PULP CANAL, DENTAL, Gates Glidden, Angle Handpiece, Corrosion-Resisting Steel Size 1,6s.
6520-01-011-2917	DRILL, PULP CANAL, DENTAL, Gates Glidden, Angle Handpiece, Corrosion-Resisting Steel, Size 2,6s.
6520-01-011-2918	DRILL, PULP CANAL, DENTAL, Gates Glidden, Angle Handpiece, Corrosion-Resisting Steel, Size 3,6s.
6520-01-011-2919	DRILL, PULP CANAL, DENTAL, Gates Glidden, Angle Handpiece, Corrosion-Resisting Steel, Size 4,6s.
6520-01-011-2920	DRILL, PULP CANAL, DENTAL, Gates Glidden, Angle Handpiece, Corrosion-Resisting Steel, Size 5,6s.
6520-01-011-2921	DRILL, PULP CANAL, DENTAL, Gates Glidden, Angle Handpiece, Corrosion-Resisting Steel, Size 6,6s.
6520-01-120-5008	DRILL, PULP CANAL, DENTAL, Peeso, Latch, Angle Handpiece, Tungsten Steel, Size 4,6s.
6520-01-120-5009	DRILL, PULP CANAL, DENTAL, Peeso, Latch, Angle Handpiece, Tungsten Steel, Size 5,6s.

6.3 Subject term (key word) listing.

Angle Handpiece
Drill, Dental
Gates Glidden
Peeso
Pulp Canal

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MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force - 03

Review Activities:

Army-MD
Navy-MS
Air Force - 03

Civil Agency Coordinating Activity:
VA-OSS

05780

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1796

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-37775B	DRILL, PULP CANAL, DENTAL (Angle Handpiece)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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